DEMONSTRATION ON OPEN DATA · OCTOBER 2026

Desk-based habitat pre-map: Fenwick Solar Farm

Pre-survey habitat information for biodiversity net gain, from open satellite data and LiDAR

SITE	Fenwick Solar Farm (EN010152), north of Doncaster, South Yorkshire
GRID REFERENCE	SE 6055 1631
AREA	Solar site, 406.6 ha, from the published Order limits
PERIOD	Survey period February 2023 to August 2024; record from 2016/17 to 2025/26
PREPARED BY	Ana Silva · kirjawa.github.io
VERSION	1, October 2026

Status. A demonstration on open data. It is not a field survey and has not been signed off by an ecologist. It does not use or reproduce the applicant's survey plan. A satellite pre-map is an input for the ecologist, never a survey result.

Summary

- **Broad habitats:** 75% cropland and 20% grassland, against 71% and 26% in the published survey. Another 3.5% is marked unclear, for the walkover.
- **A second opinion:** the Crop Map of England gives the same answer, cropland or grassland, on 93.3% of 390 ha.
- **Hedgerows from LiDAR:** 17.4 km inside the site and 29.1 km with the boundary hedges, against 28.04 km in the survey.
- **The past:** about 19 ha that had been grass for at least three years were ploughed in September 2023, inside the survey period. The Crop Map of England shows grass there in 2021, 2022 and 2023, and cereals in 2024. These are the first fields to check against the 30 January 2020 rule.
- **Water:** about 5 ha inside the site held open water in at least three of ten winters, mostly along the River Went. The record follows winter rainfall.
- **Ready for the metric:** 142 parcels laid out in the order of the statutory metric's baseline inputs, with areas that add up exactly to the site.

1 Introduction

1.1 Purpose

To show what a desk-based pre-map can give an ecologist before a biodiversity net gain survey: the broad habitats, where to look first, and what has changed on the land since 30 January 2020. Where habitat was degraded after that date, the baseline has to be the habitat before the change. This demonstration follows the order of the baseline section of a biodiversity net gain report.

1.2 Site

Fenwick Solar Farm was consented on 18 February 2026 (Fenwick Solar Farm Order 2026, SI 2026/151). The solar site studied here is the main block of the published Order limits (406.6 ha of 513 ha), separated from the cable corridor; the assessment gives about 407 ha for it. The land is low-lying, in the Humberhead Levels, with the River Went along its northern edge.

1.3 Reference survey

The applicant's Biodiversity Net Gain Assessment (AECOM, October 2024) reports the surveyed baseline in its Table F1: 421.17 ha over the solar site and a 14 ha cable corridor, surveyed from February 2023 to August 2024. The comparison here uses the shares of broad habitat types in that table. The survey's own plan is not open for reuse, so it is not used.

2 Data and methods

2.1 Data

SOURCE	USED FOR	LICENCE
Sentinel-2 L2A, 2017 to 2026	Pre-map and yearly record	Copernicus free and open
Sentinel-1 RTC, every winter pass 2016 to 2026	Open water through cloud	Copernicus free and open
LiDAR Composite 1 m: DTM, first- and last-return DSM (EA)	Hedgerows, trees, woodland, roofs	Open Government Licence
Recorded Flood Outlines (EA)	Checking radar flood maps	Open Government Licence
Rain gauges at South Elmsall, Nutwell, Dirtness (EA)	Wet and dry seasons	Open Government Licence
ERA5-Land (Copernicus Climate Change Service)	Frost and snow at each radar pass	Copernicus licence
Crop Map of England 2020 to 2024 (RPA)	Second opinion, cropland or grassland	Open Government Licence
Order limits (Planning Data platform)	Site boundary	Open Government Licence
BNG Assessment, Table F1 (AECOM, 2024)	Surveyed baseline, by shares	Published document, cited

2.2 Methods in brief

- **Pre-map.** Fields are found as areas of similar seasonal greenness, with LiDAR hedgerows as breaks, and each is classed from its own record: cropland where bare soil shows in at least two months, grassland where it stays green, woodland where LiDAR finds tall vegetation in blocks, otherwise unclear. The rules were written before the comparison and not tuned to it.
- **Hedgerows and trees.** Woody vegetation over 0.8 m that is not a roof, narrower than about 20 m, measured along its centre line in runs of 20 m or more.
- **History.** The same rules for each farming year since 2017/18. In summers at least 25% drier than usual, browned grass is not taken for bare soil.
- **Water.** Open water where both radar polarisations are dark; passes with frost or snow skipped; Sentinel-2 as the referee where it had a clear view within two days.

3 Baseline habitats

3.1 Broad habitat types

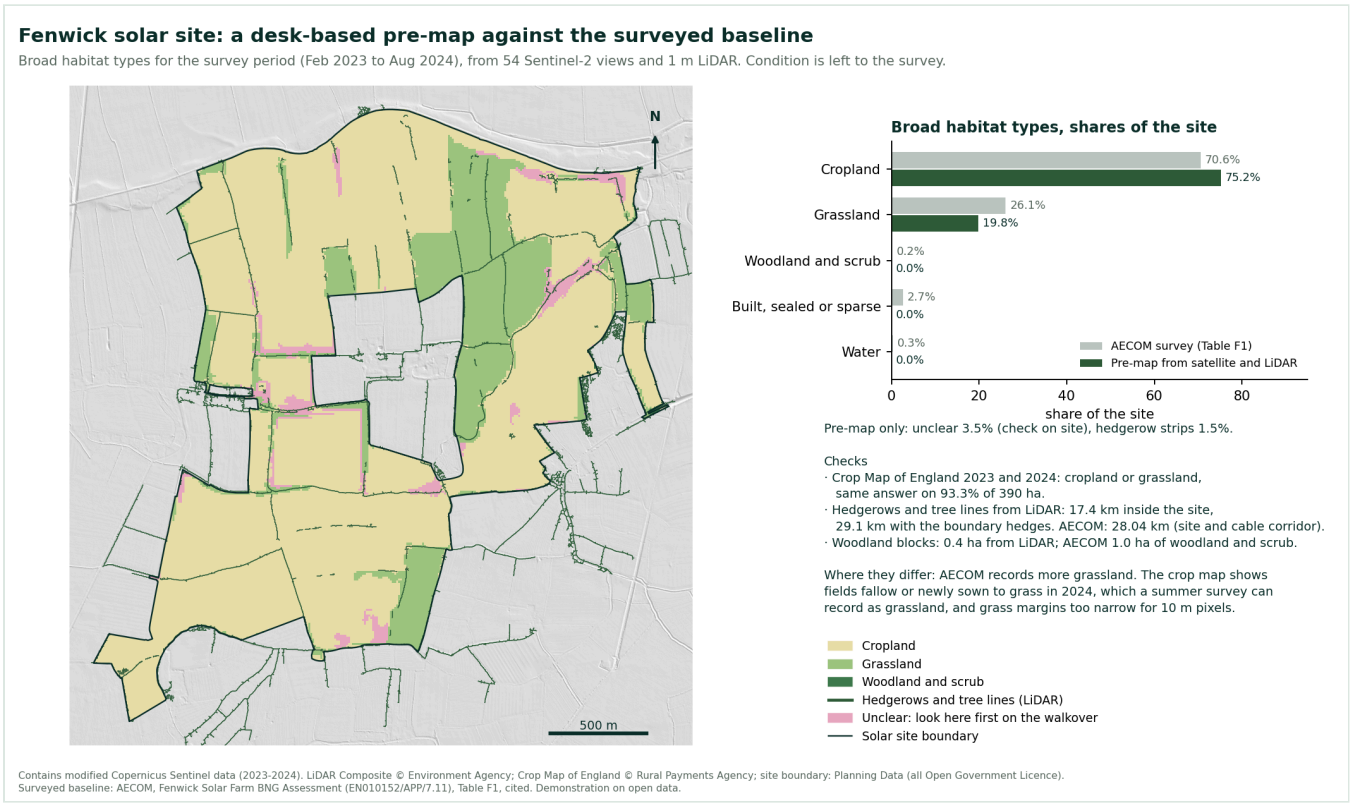


Figure 1. The pre-map of the solar site for the survey period, against the published survey's shares.

BROAD HABITAT	PRE-MAP	SURVEY (TABLE F1)
Cropland	75.2%	70.6%
Grassland	19.8%	26.1%
Woodland and scrub	0.0%	0.2%
Built, sealed or sparse	0.0%	2.7%
Water	0.0%	0.3%
Unclear, for the walkover	3.5%	n/a
Hedgerow strips (counted in km, section 3.2)	1.5%	n/a

Against the Crop Map of England for 2023 and 2024, hexagon by hexagon, the pre-map gives the same answer, cropland or grassland, on 93.3% of 390 ha. The survey records more grassland than the pre-map. The crop map shows fields left fallow or sown to temporary grass in 2024, which a summer survey can record as grassland, and grass margins narrower than 10 m are missed by the pre-map. Tracks, yards and ditches, about 3% of the site in the survey, are below what 10 m pixels can map.

3.2 Hedgerows and trees

MEASURE (LIDAR)	VALUE	SURVEY
Hedgerows and tree lines inside the site	17.4 km	
The same, on the boundary line	24.0 km	
The same, with the boundary hedges	29.1 km	28.04 km (site and cable corridor)
Woodland blocks	0.4 ha	1.0 ha of woodland and scrub
Tree crowns over 5 m, outside woodland	6.5 ha	9.3 ha of individual trees (metric areas per tree size, not like for like)
Roofs	0.06 ha	

Gaps are not bridged, so the lengths are of hedge actually standing. The LiDAR composite mixes survey years, mostly 2016 to 2022 here: a hedge planted or removed since then is not reflected. A first version of this measure treated the trees in the hedges as woodland and found 8 km; the survey's 28 km showed something was missing.

3.3 Parcel table

142 parcels, cut along the hedgerows and the pre-map's class boundaries, laid out in the order of the statutory metric's on-site habitat baseline inputs. Broad habitat and area are filled in, habitat type is suggested, and condition, irreplaceability and strategic significance are left to the survey and the Local Nature Recovery Strategy. Areas are rounded to 0.01 ha so that they add up to the site's 406.57 ha. The full table is in the download.

BROAD HABITAT	PARCELS	AREA (HA)
Cropland	28	309.47
Grassland	72	82.57
Unclear (walkover)	39	14.39
Urban	1	0.02
Woodland and forest	2	0.12

The largest parcels of each type:

REF	BROAD HABITAT	HABITAT TYPE	AREA (HA)	EVIDENCE
P001	Cropland	Cereal crops or other arable (to confirm)	98.95	bare soil in about 7 months of Feb 2023 to Aug 2024
P002	Cropland	Cereal crops or other arable (to confirm)	87.47	bare soil in about 4 months of Feb 2023 to Aug 2024; open water in winter on 4.50 ha, in up to 10 of 10 winters
P003	Cropland	Cereal crops or other arable (to confirm)	40.81	bare soil in about 5 months of Feb 2023 to Aug 2024
P004	Cropland	Cereal crops or other arable (to confirm)	23.93	bare soil in about 3 months of Feb 2023 to Aug 2024
P029	Grassland	Modified or other neutral grassland (to confirm)	39.96	green all along (NDVI 10th percentile 0.58); open water in winter on 0.19 ha, in up to 7 of 10 winters
P030	Grassland	Modified or other neutral grassland (to confirm)	10.29	green all along (NDVI 10th percentile 0.39)
P031	Grassland	Modified or other neutral grassland (to confirm)	9.18	green all along (NDVI 10th percentile 0.64)
P032	Grassland	Modified or other neutral grassland (to confirm)	5.02	green all along (NDVI 10th percentile 0.44)
P101	Woodland and forest	To confirm: woodland or scrub	0.07	tall vegetation in a block (LiDAR)
P102	Woodland and forest	To confirm: woodland or scrub	0.05	tall vegetation in a block (LiDAR)
P103	Urban	Developed land; sealed surface (to confirm)	0.02	a roof (LiDAR: first and last returns together)
P104		Unclear from satellite: check on the walkover	2.53	mixed or changing signal
P105		Unclear from satellite: check on the walkover	1.44	mixed or changing signal
P106		Unclear from satellite: check on the walkover	1.36	mixed or changing signal
P107		Unclear from satellite: check on the walkover	0.97	mixed or changing signal

4 Site history and the 30 January 2020 cut-off

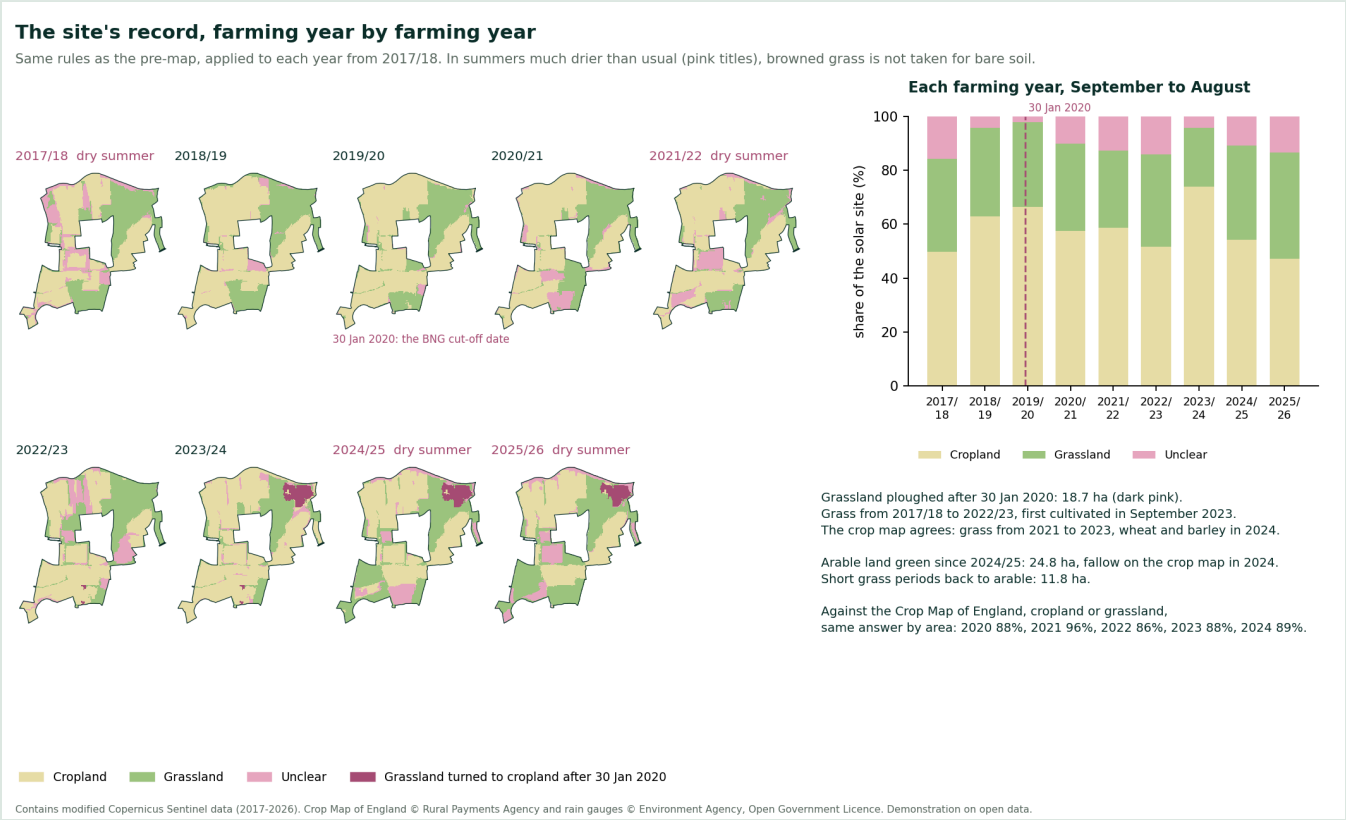


Figure 2. The record of the solar site, farming year by farming year, with grassland ploughed after 30 January 2020 in dark pink.

- **Grassland ploughed after the cut-off: 18.7 ha.** Grass in the record from 2017/18 to 2022/23, first cultivated in September 2023, inside the survey period. The Crop Map of England shows grass there in 2021, 2022 and 2023, and winter wheat and spring cereals in 2024. Where habitat was degraded after 30 January 2020, the baseline has to be the habitat before the change, so these are the first fields to check. The survey's own map isn't open, so how it recorded these fields isn't known here.
- **Arable land green since 2024/25: 24.8 ha,** fallow on the crop map in 2024. Resting arable, which a survey in 2025 or 2026 could record as grassland.
- **Short grass periods back to arable: 11.8 ha,** more likely temporary leys, which UKHab counts as cropland.
- **Check against the Crop Map of England,** cropland or grassland, same answer by area: 2020: 88%, 2021: 96%, 2022: 86%, 2023: 88%, 2024: 89%.

5 Water and weather

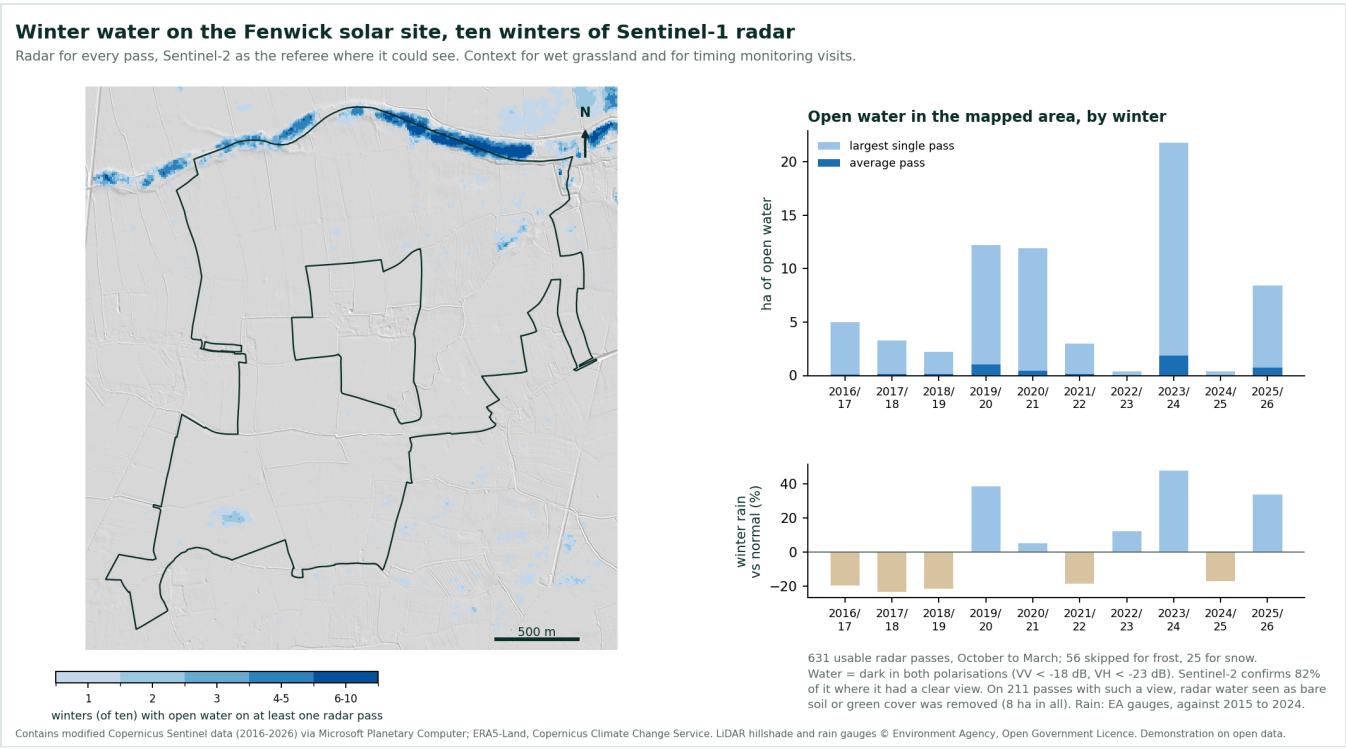


Figure 3. Open water in winter over ten winters, with winter rainfall at the Environment Agency's rain gauges.

- **The record.** 631 usable winter radar passes; 56 skipped for frost and 25 for snow, which darken radar as water does.
- **Checked against Sentinel-2.** The first version, dark in one polarisation only, did not follow the weather: the dry winter 2018/19 came out the wettest. Where a clear Sentinel-2 view existed within two days, only 9% of its radar water was water and 82% was green crops and grass. With both polarisations dark, 82% is confirmed as water, and on 211 passes radar water that the optical view showed as bare soil or green cover was removed.
- **Where water stands.** About 5 ha inside the site held open water in at least three of ten winters, mostly along the River Went: the place to look for floodplain grassland and wetland, which the survey records over 8.2 ha.
- **Flood events.** On 13 November 2019, 80% of the open water radar saw lies inside the EA's outlines from aerial photography. On 22 January 2021, after Storm Christoph, radar saw water over half of the EA's outlines and along the River Went's floodplain, where the EA drew no outline for that storm.
- **Weather during the survey.** Winter 2023/24 was the wettest of the record (+48% rain), and the summers of 2022 and 2025 the driest: context for how wet the ground looked when it was surveyed, and for the timing of monitoring visits.

6 Where to look first on the walkover

- The 14.4 ha marked unclear in the parcel table.
- The 19 ha of grassland ploughed in September 2023, for the 30 January 2020 rule.
- The wet margin along the River Went, for floodplain grassland and wetland.
- The 25 ha of fallow arable since 2024, which can pass for grassland.
- Hedgerow trees (6.5 ha of crowns from LiDAR), counted as individual trees in the metric.
- Grass margins and ditches, too narrow for the pre-map.

7 What satellite data can't tell here

- **Condition,** which needs the field survey and the condition criteria.
- **Finer types:** other neutral and modified grassland look alike from space.
- **Small and narrow habitats:** ditches, tracks, yards and grass margins under 10 m wide.
- **Species,** and anything that depends on them.
- **The date of the LiDAR,** a composite of survey years.
- **Water under crops or trees,** and flooding that drained between radar passes.

8 How AI was used

An agent (Claude Code) wrote the scripts: downloads from the open services, the pre-map and parcels, the LiDAR measures, the flood checks, the figures and this report. Every result was checked against something independent: the published survey totals, the Crop Map of England, the Environment Agency's flood outlines and Sentinel-2. What it got wrong, and what caught it: the first hedgerow measure (8 km, against the survey's 28 km); the first winter water record, which followed green crops rather than water (caught by the weather and by Sentinel-2); and a crop map download that covered only part of the site (caught by a gap in the history check).

9 Sources

- AECOM (2024). Fenwick Solar Farm Biodiversity Net Gain Assessment, EN010152/APP/7.11, Table F1.
- Defra (2024, updated 2026). Statutory biodiversity metric tools and guides; What to include in biodiversity net gain baselines for NSIPs.
- CIEEM (2021). Biodiversity Net Gain report and audit templates.
- Planning Data platform, infrastructure-project dataset (Order limits). Rural Payments Agency, Crop Map of England (CROME) 2020 to 2024.
- Environment Agency: LiDAR Composite (1 m), Recorded Flood Outlines, Hydrology rain gauges. Copernicus Sentinel-1 and Sentinel-2.
- Muñoz Sabater, J. et al. (2021). ERA5-Land. Earth System Science Data 13. Twele, A. et al. (2016). Sentinel-1-based flood mapping. International Journal of Remote Sensing 37.

Contains modified Copernicus Sentinel data (2016 to 2026). Environment Agency, Rural Payments Agency and Planning Data content under the Open Government Licence. ERA5-Land generated using Copernicus Climate Change Service information (2026).